Dr. Ashish Singh

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Email: ashishnit07@gmail.com, ashish@iiitp.ac.in

Address: ECE department, IIIT Pune, Maharashtra, India

Career Objective: Seeking for a career that is interesting and challenging to achieve high career growth through continuous process of learning, with a reputed institute where, I could use my knowledge and skills for the mutual growth and benefit.

EDUCATION

National Institute of Technology (NIT) Hamirpur, Himachal Pradesh, India
Ph.D. in Low Power VLSI Interconnects
Dissertation Title: "*Study of Mixed Carbon Nanotube Bundles for VLSI Interconnects in Sub-threshold Regime*".

Thapar University, Patiala, Punjab, India
M.Tech in VLSI Design
Dissertation Title: "*Test time optimization of SOCs using Heuristic algorithms*".
CGPA: 7.85

Uttar Pradesh Technical University (UPTU), Lucknow, Uttar Pradesh, India
B.Tech in Electronics and Communication Engineering
Project Title: "*Telephone based auto muting light ON circuit with call record*".
Percentage: 74.80%

Lucknow Public Inter College, Lucknow, Uttar Pradesh, India
Intermediate with Honors from U.P Board
Percentage: 81.40%

Lucknow Public Inter College, Lucknow
High School with Honors from U.P Board
Percentage: 77.50%

TEACHING EXPERIENCE

Indian Institute of Information Technology (IIIT) Pune — Maharashtra, India
Adjunct Assistant Professor in Dept. of Electronics & Communication Engineering — [Jan 2026 – Present]

Punjab Engineering College (Deemed to be University), Chandigarh, India — Chandigarh, India
Temporary Faculty in Dept. of Electronics & Communication Engineering — [Aug 2025 – Dec 2025]

Indraprastha Institute of Information Technology (IIIT) Delhi — Delhi, India
Teaching Faculty in Dept. of Electronics & Communication Engineering — [Aug 2022 – Oct 2022]

National Institute of Technology (NIT) Kurukshetra, — Haryana, India
Temporary Faculty in School of VLSI Design & Embedded System — [Nov 2020 –Dec 2021]

RESEARCH PUBLICATIONS

- ### International Journals

[1] Ajay Kumar, **Ashish Singh** and Amit Kumar, "Signal Integrity Analysis of Cu-CNT Based Shield DM-TGV Using Temperature and Roughness-Aware EM-RA Technique for 3D Integration," ***IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems,*** vol. xx, pp. xx–xx, 26th Dec 2025 (*in early access*). **DOI: 10.1109/TCAD.2025.3648614.**

[2] **Ashish Singh** and Rohit Dhiman, "Proposal and Analysis of Mixed CNT Bundle for Sub-Threshold Interconnects," ***IEEE Transactions on Nanotechnology***, vol. 18, pp. 584–588, 4th June 2019. **DOI: 10.1109/TNANO.2019.2919445.**

[3] **Ashish Singh**, Brajesh Kumar Kaushik and Rohit Dhiman, "Modeling and Analysis of Cu-Carbon Nanotube Composites for Sub-Threshold Interconnects," ***IEEE Open Journal of Nanotechnology,*** vol. 3, pp. 236–243, 10th November 2022. **DOI: 10.1109/OJNANO.2022.3221141.**

[4] **Ashish Singh**, Rajeevan Chandel, and Rohit Dhiman, "Proposal and analysis of relative stability in mixed CNT bundle for sub-threshold interconnects," ***INTEGRATION, the VLSI Journal*** **(Elsevier)**, vol. 80, pp. 29–40, September 2021. **DOI**: **https://doi.org/10.1016/j.vlsi.2021.05.004**.

[5] Ajay Kumar, Amit Kumar, and **Ashish Singh***, "Frequency and Time-Domain Analysis of Shield Differential Multibit TGV Using EM-RA Technique for 3D ICs," ***Micro Electronics Journal*** **(Elsevier),** vol. 126, pp. 1–14, 11th December 2025. **DOI: https://doi.org/10.1016/j.mejo.2025.107014**.

[6] **Ashish Singh,** Ajay Kumar and Amit Kumar, "Signal Integrity Analysis in Mixed CNT Bundle Interconnects using EM-RA," ***Journal of Computational Electronics*** **(Springer),** vol. 24, pp. 1–10, 21st June 2025. **DOI: https://doi.org/10.1007/s10825-025-02353-y.**

[7] Amit Kumar, Ajay Kumar and **Ashish Singh***, "Proposal and Analysis of Shield Differential Multibit TGVs for 3D Integration," ***IEEE Transactions on Electromagnetic Compatibility.*** [**Status**: *Communicated* on 5th Jan 26].

- ### International Conferences

[1] **Ashish Singh**, Rohit Dhiman, and Rajeevan Chandel, "Modeling of Mixed CNT bundle for Sub-threshold Interconnects," ***IEEE Electrical Design of Advanced Packaging and Systems Symposium (EDAPS)***, held at Taj Chandigarh, India, 16th–18th Dec 2018. **DOI: 10.1109/EDAPS.2018.8680870.** [**Scopus Indexed**].

[2] **Ashish Singh,** et al., "Delay and frequency investigations in coupled MLGNR interconnects," *in Proceedings*, ***21st International Symposium on VLSI Design and Test*** **(VDAT)**, held at IIT Roorkee, India, pp. 429–440, 29th June–2nd July 2017. **DOI: https://doi.org/10.1007/978-981-10-7470-7_43.** [**Scopus Indexed**].

AREA OF RESEARCH

- Modeling of *Carbon nanotube* (*CNT*), *Graphene-nanoribbon*, and *Copper-CNT*-based VLSI interconnect.
- Through-silicon vias (TSVs), through-glass via (TGV) and, 3D-dimensional IC packaging.
- *Ultra-low power* and *high-speed VLSI interconnect*s circuit design for different applications.
- Nano-electronics, and semiconductor technology.

COMPETATIVE EXAMS QUALIFIED

- **National Eligibility Test (UGC-NET)–Assistant Professor** in Electronic Science in 2016.
- **Graduate Aptitude Test in Engineering (GATE)** in Electronics & Comm. Eng. in 2014.
- **All India Talent Search Examination** at *National Level* and was awarded '*Certificate of merit*' in 2006.

HONORS AND AWARDS

- Awardee, **SERB-DST Fellowship,** Govt. of India, New Delhi during Ph.D.
- Awardee, **GATE Fellowship**, **MHRD**, Govt. of India, New Delhi during M.Tech.
- Awarded **'Certificate of Honor'** for scoring Distinction in *High School* and *Intermediate U.P. Board* exam.

Publications Homepage:

- Google scholar: (https://scholar.google.com/citations?hl=en&authuser=1&user=uTra3wcAAAAJ)
- ResearchGate: (https://www.researchgate.net/profile/Ashish-Singh-87)
- ORCID Id: 0000-0002-6057-5362

I hereby declare that the information given herein are true & complete to the best of my knowledge and belief.

Place: Pune, Maharashtra **Dr. Ashish Singh**